

555.644
Assignment 6

Problem 1 (7.1)

Companies A and B have been offered the following rates per annum on a \$20 million five-year loan:

	<i>Fixed Rate</i>	<i>Floating Rate</i>
Company A	5.0%	SOFR+0.1%
Company B	6.4%	SOFR+0.6%

Company A requires a floating-rate loan; company B requires a fixed-rate loan. Design a swap that will net a bank, acting as intermediary, 0.1% per annum and that will appear equally attractive to both companies.

Problem 3 (7.4)

A currency swap has a remaining life of 15 months. It involves exchanging interest at 10% on £20 million for interest at 6% on \$30 million once a year. The term structure of interest rates in both the United Kingdom and the United States is currently flat, and if the swap were negotiated today the interest rates exchanged would be 4% in dollars and 7% in sterling. All interest rates are quoted with annual compounding. The current exchange rate (dollars per pound sterling) is 1.5500. What is the value of the swap to the party paying sterling? What is the value of the swap to the party paying dollars?

Problem 3 (7.5)

Explain the difference between the credit risk and the market risk in a financial contract.

Problem 4 (7.8)

Companies X and Y have been offered the following rates per annum on a \$5 million 10-year investment:

	<i>Fixed Rate</i>	<i>Floating Rate</i>
Company X	8.0%	SOFR
Company Y	8.8%	SOFR

Company X requires a fixed-rate investment; company Y requires a floating-rate investment. Design a swap that will net a bank, acting as intermediary, 0.2% per annum and will appear equally attractive to X and Y.

Problem 5 (7.10)

A financial institution has entered into a 10-year currency swap with company Y. Under the terms of the swap, the financial institution receives interest at 3% per annum in Swiss francs and pays interest at 8% per annum in U.S. dollars. Interest payments are exchanged once a year. The principal amounts are 7 million dollars and 10 million francs. Suppose that company Y declares bankruptcy at the end of year 6, when the exchange rate is \$0.80 per franc. What is the cost to the financial institution? Assume that, at the end of year 6, the interest rate is 3% per annum in Swiss francs and 8% per annum in U.S. dollars for all

maturities. All interest rates are quoted with annual compounding.

Problem 6

The SOFR zero curve is flat at 5% (continuously compounded) out to 1.5 years. Swap rates for 2- and 3-year semiannual pay swaps are 5.4% and 5.6%, respectively. Estimate the SOFR zero rates for maturities of 2.0, 2.5, and 3.0 years. (Assume that the 2.5-year swap rate is the average of the 2- and 3-year swap rates.)

Problem 7

Under the terms of an interest rate swap, a financial institution has agreed to pay 10% per annum and receive three-month SOFR in return on a notional principal of \$100 million with payments being exchanged every three months. The swap has a remaining life of 14 months. The average of the bid and offer fixed rates currently being swapped for three-month SOFR is 12% per annum for all maturities. The three-month SOFR rate one month ago was 11.8% per annum. All rates are compounded quarterly. What is the value of the swap?

Problem 9 (7.22)

Company A, a British manufacturer, wishes to borrow U.S. dollars at a fixed rate of interest. Company B, a US multinational, wishes to borrow sterling at a fixed rate of interest. They have been quoted the following rates per annum (adjusted for differential tax effects):

	<i>Sterling</i>	<i>US Dollars</i>
<i>Company A</i>	11.0%	7.0%
<i>Company B</i>	10.6%	6.2%

Design a swap that will net a bank, acting as intermediary, 10 basis points per annum and that will produce a gain of 15 basis points per annum for each of the two companies.